MILITARY SPECIFICATION

SPECIFICATIONS, TYPES AND FORMS; NAVORD XWS AND WS SERIES

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements for the preparation of specifications in the NAVORD XWS and WS series which describe program peculiar items, processes, and materials procurred by the Naval Ordnance Systems Command.

1.2 Classification. Specifications shall be of the following series, types, and forms as specified in the contract or order (see 6.2):

a. Series.

XWS - NAVORD Experimental-Developmental Weapons Specification (see 6.1.1a)

WS - NAVORD Weapons Specification (see 6.1.1b)

OS - NAVORD Ordnance Specification/Description and Requirements (an obsolete series no longer prepared, see 3.8 and 6.1.1c)

b. Types. As defined in MIL-S-83490 (see 3.2 and 6.2.1)

c. Forms.

Form 1a - Specifications which conform to MIL-STD-490 in all details, including specific text structure, paragraph numbering, and paragraph titling specified in the appendixes of MIL-STD-490 (see 3.3 and 6.1.2)

Form 1b - Specifications which conform to MIL-STD-490 except that specific paragraph sequencing, numbering, and titling specified in the appendixes of MIL-STD-490 need not apply. However, the section numbers and titles are mandatory (see 3.3 and 6.1.2).

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2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein. In the event of conflict between this specification and other documents referenced herein, requirements of this specification shall apply.

SPECIFICATIONS

Military

MIL-Q-9858	Quality Program Requirements
MIL-M-38748	Microfiche; for Engineering/Technical Data, Reports, Studies and Related Data, Requirements for
MIL-I-45208	Inspection System Requirements
MIL-S-83490	Specifications, Types and Forms

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-414	Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-490	Specification Practices
MIL-STD-690	Failure Rate Sampling Plans and Procedures

PUBLICATIONS

Manuals

Naval Ordnance Systems Command

OP 5	Ammunitions Ashore, Handling, Stowing and Shipping

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Data Item Descriptions

Naval Ordnance Systems Command

UDI-E-20602 Test Reports, Appendant

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issues in effect on date of invitation for bids or request for proposals shall apply. In the event of conflict between this specification and other documents referenced herein, requirements of this specification shall apply.

DoD Manual 4120.3-M	Standardization Policies, Procedures and Instructions
DoD Manual 4145.26M	DoD Contractors' Safety Manual for Ammunition, Explosives and Related Dangerous Material

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 Series of specifications. Each specification shall conform to one of the series contained herein (see 1.2a) and shall follow the outline of forms and requirements respectively prescribed herein.

3.2 Types of specifications. The types of specifications prepared shall be in accordance with MIL-S-83490 and as specified in the contract or order.

3.3 Forms of specifications. The forms of specifications prepared shall be in accordance with MIL-S-83490, form 1a or form 1b as specified in the contract or order (see 1.2c and 6.1.2).

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3.4 Specification preparation. Specifications written hereunder shall contain six numbered sections and be prepared in accordance with MIL-S-83490 and MIL-STD-490.

3.4.1 Formats. Figures 1 and 2 provide a sample format for XWS's and WS's respectively.

3.4.2 Assignment of numbers. XWS and WS specification numbers are controlled and issued by the U. S. Naval Ordnance Station, Technical Documents Department (Code 80111), Louisville, Ky. 40214. Requests for numbers shall include:

a. The name and address of the contractor or activity actually writing the specification.

b. The contract or task assignment number (as applicable).

c. The nomenclature of the equipment designated by the contract or task assignment to which the specification number(s) shall apply.

d. The quantity of numbers required.

3.4.3 Report of assignment. As specified in the contract or order, each contractor (or Naval activity as directed) using NAVORD specification numbers shall prepare a list of the numbers used (see 6.2.2).

3.4.4 Cover sheet. A cover sheet shall be prepared and attached to each specification prepared hereunder. Figure 3 is a sample cover sheet. Date of approval as shown on the cover sheet shall be the same as the date on the first page of the XWS and WS.

3.4.5 Contract number. When the specification is prepared under contract, the contract number shall be included in section 6 of the specification in accordance with the following sample entry:

"6.5 This specification was prepared by Waterhouse, Inc., and furnished to the Government under Contract No. N00156-69-C-1523."

3.4.6 Section 2 of specifications. The introductory statements in section 2 of specifications prescribed herein shall include the requirement that, "In the event of conflict between this specification and

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other documents referenced herein, requirements of this specification shall apply." All and only those documents identified and referenced in sections 3, 4, and 5 and appendixes of specifications shall be listed in section 2 (see figures 1 and 2 herein).

3.4.7 Classified specifications. Each paragraph of a classified specification shall be appropriately marked using the symbols (TS) for top secret, (S) for secret, (C) for confidential, and (U) for unclassified. The symbol shall be placed immediately preceding and to the left of the paragraph involved. If all paragraphs in a classified specification have the same classification, a statement to that affect shall be included and individual paragraphs need not be marked.

3.4.8 Changes and revisions. Specifications shall be corrected or updated by means of a specification change notice (SCN) or revision in accordance with MIL-STD-490.

3.4.9 Identification of changes. Revisions of specifications shall include asterisks in the margins of the pages to indicate where changes have been made with respect to the prior issue. The following note shall be included in the NOTES (section 6) of the specification:

"6._ Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modification, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue."

If the changes are extensive and too numerous to annotate, the following note will be included in the NOTES (section 6) of the specification:

"6._ Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes."

3.4.10 Safety precautions. Specifications covering explosive-loaded items, explosives, or propellants shall contain the following in the "ordering data" paragraph of section 6:

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"6. The safety precaution requirements of the "Contractors' Safety Manual for Ammunition, Explosives and Related Dangerous Material" (DOD 4145.26M) are applicable.

NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunitions Ashore" (OP 5) should be made applicable."

3.5 Stipulation of requirements and inspections.

3.5.1 Sections 3 and 5. The requirements covered in sections 3 and 5 of a specification shall indicate, as definitely as practicable, the minimum standard of quality, workmanship, and performance requirements which the commodity or service, and preparation for delivery thereof, must meet to be acceptable. The requirements should be so worded so as to provide a definite basis for rejection in the cases where the quality, workmanship, or performance are such that the item is unsuitable for the purposes intended. However, requirements should not be specified to an unnecessary degree of exactness or restrictiveness, but be detailed only to the degree necessary to ensure the procurement of items adequate for the purpose intended.

3.5.1.1 Use of requirements stated in other documents. Requirements stated in other documents (specifications, standards, etc.) may be used and referenced in specifications prepared hereunder. However, the extent to which the requirements apply should be clearly indicated. Only documents which can readily be obtained by the specification users should be referenced. The appendix hereof lists a number of documents which have either been specifically prepared for use in weapons development or have been thoroughly reviewed and are considered suitable for use in weapons development.

3.5.2 Section 4. This section shall include all of the examinations and tests (by reference where applicable) to be performed in order to ascertain that the product, material, or process to be developed or offered for acceptance conforms to all the requirements listed in sections 3 and 5 of the specification. If this is not done, the obligation of the contractor and the prerogative of the Government to validate item performance may be abrogated. Conversely, section 4 shall not impose any inspection which does not correspond to a requirement listed in section 3 or 5. In short, there shall be a one-to-one correspondence between requirements and inspections.

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3.5.2.1 Qualification. Product qualification provisions as provided for in chapter 4 of the Defense Standardization Manual 4120.3-M shall not be utilized in XWS, WS, or OS specifications. Qualified products lists shall not be established against specifications prepared hereunder.

3.5.2.2 Quality conformance inspection sampling. Standard sampling procedures in accordance with MIL-STD-105, MIL-STD-414, MIL-STD-690, or other applicable standards shall be prescribed in specifications in preference to unique sampling procedures. Note that acceptable quality levels (AQL's) and other details from such standards must be stated. In certain cases 100 percent inspection may be required in lieu of sampling.

3.5.2.3 Contract quality requirements. MIL-I-45208 and MIL-Q-9858 shall not be referenced in commodity specifications. Inspection system and quality program requirements are contractual considerations and their inclusion in specifications is prohibited by MIL-STD-490.

3.5.2.4 Appendix report form. A preprinted report form for recording results of quality conformance examinations and tests may be included as an appendix to a specification. Its use shall be limited to specifications covering complex items involving extensive testing of products. These appendixes, when used, shall conform to the sample format shown in figure 4. The report form shall be specified in sections 4 and 6 of the specification as indicated herein, citing Data Item Description (DID) UDI-E-20602 as the approved DID for acquiring the completed form. Each specification which includes such an appendix shall specify in section 4 that the appendix may be duplicated without restriction for use by testing activities.

3.6 Data requirements. Where applicable, provisions for procurement or delivery of data may be included in specifications as indicated in MIL-STD-490. These contract data requirements are to be included under the heading of "6.2 Ordering Data" of the specification, segregated into "Procurement Requirements" and "Contract Data Requirements." Under the "Contract Data Requirements" subheading, identify the requirements for deliverable data in other sections of the specification in the manner illustrated by 6.2 of this specification.

3.7 Conversion of an XWS to a WS. When an XWS specification has served its intended purpose, and when further development or procurement of the specified item for service use is required, the XWS may be

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converted to a WS. In doing this, the "X" and the suffix (if any) are deleted and the specification is then identified by the WS symbol with the same sequential arabic serial number. Additionally, the new date of approval and supersession data are put in the upper right corner. The first issue of the WS will be considered a revision of the XWS, thus being assigned the next appropriate letter.

3.8 Revision of Ordnance Specifications/Description and Requirements (OS's). If an OS needs revision in its entirety, one of the following alternatives shall be utilized as specified in the contract or order:

(a) Issue the specification in the WS series, assigning it a number from the WS series

(b) Reissue the specification in the OS series.

3.9 Microfiche requirements. Microfiche of each approved specification prepared hereunder shall be furnished as specified in the contract or order (see 6.2.1 and 6.2.3). The microfiche shall conform to type I, class 1, style 2 of MIL-M-38748. Actual display of letters and numbers for grid area row and image identification on the microfiche is optional. However, if display is desired, the following exception to MIL-M-38748 shall apply: rows of images shall be lettered vertically top to bottom, A through G, (title area, described as row A in MIL-M-38748 will not have a letter designation).

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance provisions shall be as specified in MIL-S-83490.

5. PREPARATION FOR DELIVERY

5.1 Preparation of data for delivery shall be as specified in MIL-S-83490.

6. NOTES

6.1 Intended use. Specifications acquired hereunder are intended for use in original procurement of program peculiar items, processes, and materials by or for the Naval Ordnance Systems Command.

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6.1.1 Series. The intended use of each specification series is as follows:

a. XWS's - NAVORD Experimental-Developmental Weapons Specifications:

(1) to state the design criteria and performance requirements to be used as goals in a development program for an equipment, weapon, or weapon system.

(2) to state requirements and quality assurance provisions to be used in lieu of military series specifications for procurement of a limited number of items incident to research and development, or procurement of items for test and evaluation.

b. WS's - NAVORD Weapons Specifications:

(1) for procurement of items which are peculiar to requirements of the Naval Ordnance Systems Command

(2) for the purchase of items required at the Naval Ordnance field activity level which do not warrant central control within the Naval Ordnance Systems Command and do not warrant the preparation of a Military or Federal specification.

c. OS's - NAVORD Ordnance Specifications/Description and Requirements:

(1) for the same purposes as XWS and WS specifications. However, since the OS series is obsolete, if an OS requires revision it should be issued in the WS series, unless it is essential to retain it in the OS series for logistics management or configuration control purposes.

6.1.2 Forms. The intended use of each specification form is as follows:

a. Form 1a - These specifications are prepared to rigid military standards and will be ordered only when the Government's intended use requires controls inherent to such standards including extensive control of paragraph content and format. For example, this form is intended for use where computer programs require strict uniformity of paragraph numbering and specification structure.

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b. Form 1b - These specifications are prepared to military standards and will normally be ordered when the Government's intended use requires maximum control of the specification content with limited format control. For example, this form is used for XWS and WS specifications if the rigid paragraph numbering and specification structure controls are unnecessary.

6.2 Ordering data. DD Form 1423 and other related procurement documents should specify the following:

6.2.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Series of specification; XWS, WS, or OS (see 3.1, 3.7, 3.8 and 6.1.1).

c. Type of specification (see 3.2).

d. Form of specification (1a or 1b). Form 1b shall normally be specified (see 1.2, 3.3, and 6.1.2).

e. Assignment of specification number if different than 3.4.2.

f. Reproducible and nonreproducible copies required (including microfiche per 3.9) (see 6.2.3 and 6.2.4).

g. Requirements for preparation for delivery (see MIL-S-83490).

6.2.2 Contract data requirements. Items of deliverable data required by this specification are cited in the following paragraphs:

Paragraph	Data requirement	Applicable DID
1.2 and 3.1	Specifications, NAVORD Weapons (WS Series)	UDI-E-20600
1.2 and 3.1	Specifications, NAVORD Experimental-Developmental Weapons (XWS Series)	UDI-E-20599
3.4.3	Report, Status, Specification (For assignment of XWS and WS numbers)	UDI-E-20603

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The DID's (Data Item Descriptions/DD Forms 1664) for the above data requirements are approved by the Naval Ordnance Systems Command for interim use pending inclusion in the Department of Defense Authorized Data List (TD-3). Such data will be delivered as identified on the above DID's when specified on DD Form 1423 (Contract Data Requirements Lists) and incorporated into applicable contracts (to which the DID's should be physically attached). Copies of these DID's may be obtained from the Commanding Officer, Naval Ordnance Station, Standardization Division (502), Indian Head, Md. 20640.

6.2.3 Copies for data repository. DD Form 1423 should indicate that, within 30 days of specification authentication by the activity delegated approval authority, one microfiche copy (see 3.9) shall be forwarded to:

Commanding Officer
Naval Ordnance Station
Technical Documents Dept. (Code 80111)
Louisville, Ky. 40214

6.2.4 Copies of specifications. When various reproducible and non-reproducible copies of specifications are required for delivery in addition to the microfiche repository copy indicated in 6.2.3, MIL-D-5480 should be cited on the DD Form 1423 to the extent appropriate.

Preparing activity:
Navy - OS
Project No. MISC-N978-1

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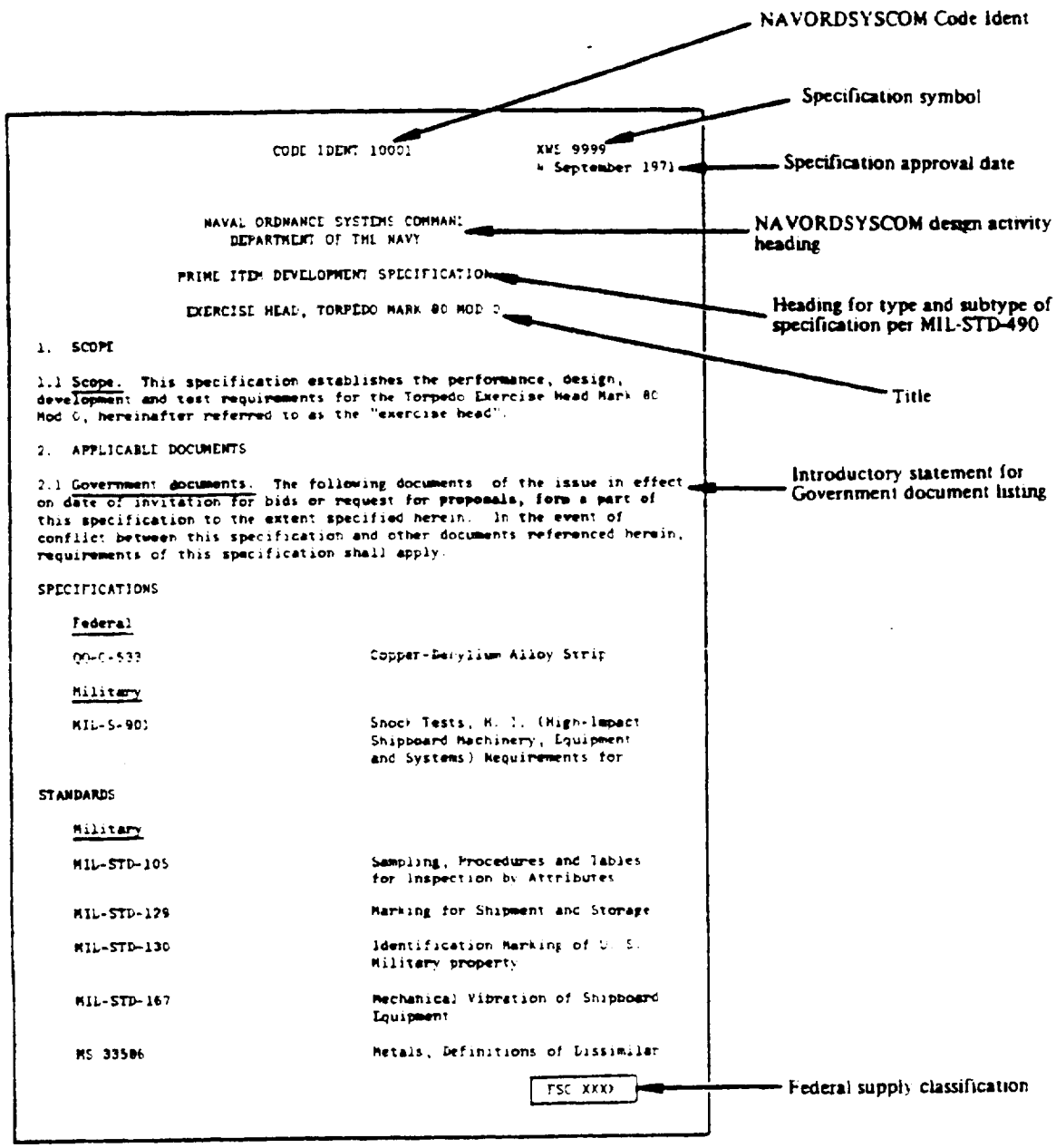


FIGURE 1. SAMPLE FORMAT FOR AN XWS SPECIFICATION

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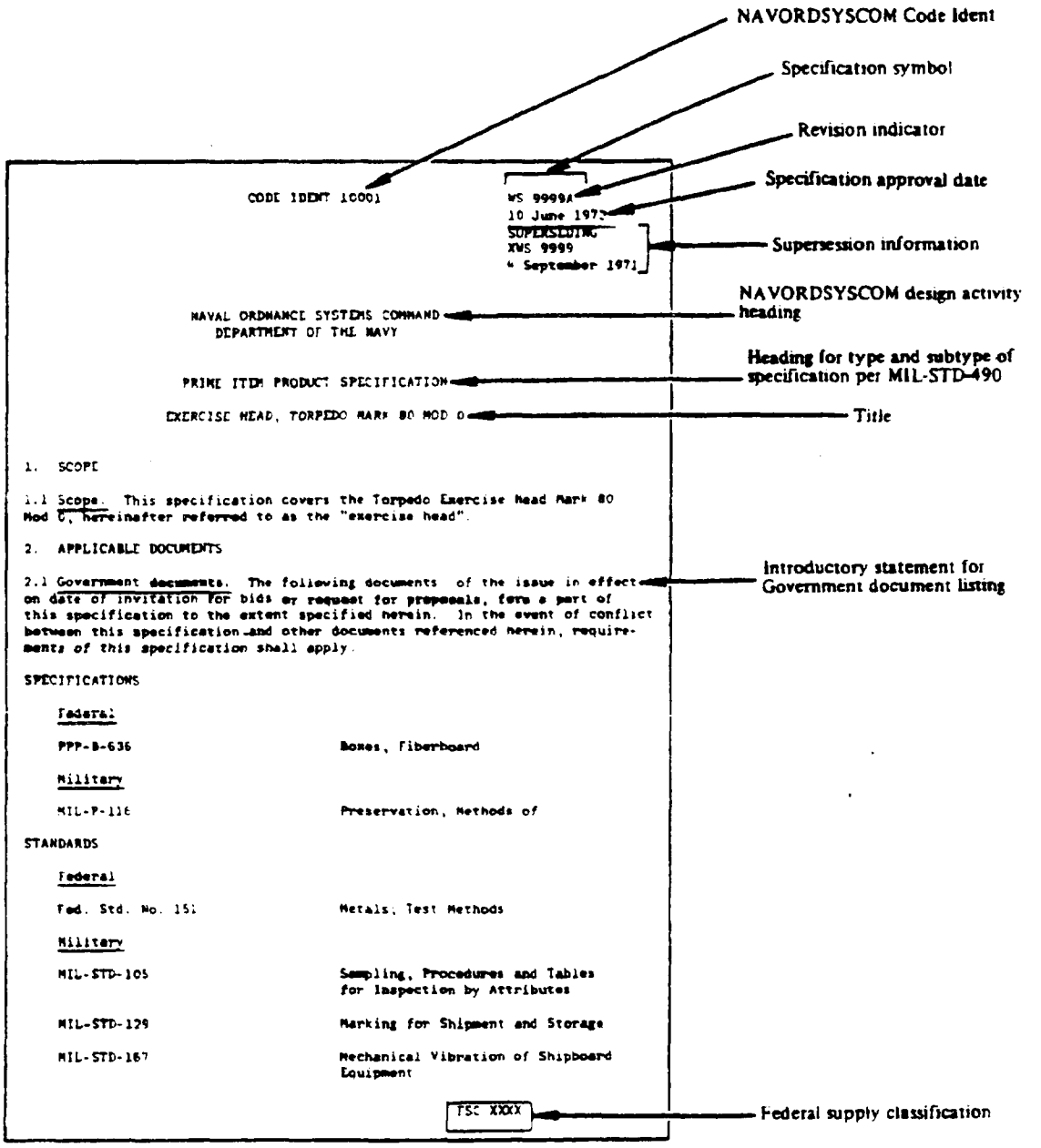


FIGURE 2. SAMPLE FORMAT FOR A WS SPECIFICATION

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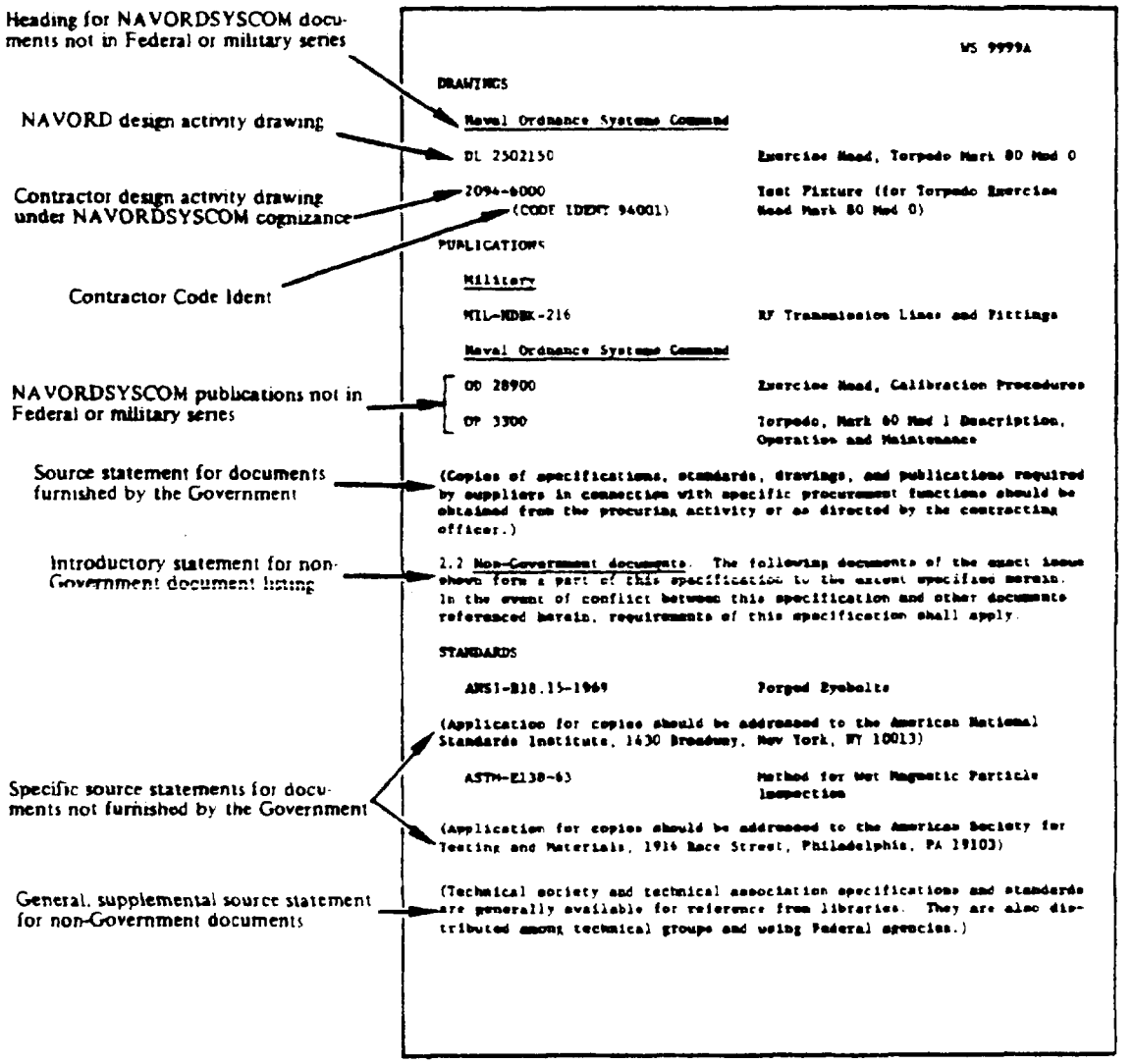


FIGURE 2.(CONTD) SAMPLE FORMAT FOR A WS SPECIFICATION

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CODE IDENT 10001

WS 9999A
10 June 1972
SUPERSEDING
XWS 9999
4 September 1971

NAVAL ORDNANCE SYSTEMS COMMAND
DEPARTMENT OF THE NAVY
PRIME ITEM PRODUCT SPECIFICATION

EXERCISE HEAD, TORPEDO MARK 80 MOD 0

TRAJECTORY DIVISION
THE BALLISTICS CORPORATION
ORDNANCE, INDIANA

APPROVED _____ DATE _____

PREPARED		
REVIEWED		
PROJECT ENGINEER		

APPROVED FOR NAVORD

By direction of the Commander
Naval Ordnance Systems Command

19 _____

FSC XXXX

FIGURE 3. SAMPLE COVER SHEET

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WS 9998
APPENDIX

REPORT FORM

QUALITY ASSURANCE PROVISION

QUALITY CONFORMANCE TESTS AND EXAMINATIONS

FOR

INSTRUMENTATION ASSEMBLY

DL 2539153 (Type I), DL 2539071 (Type II),

or DL 2539544 (Type III)

(For Exercise Head Mk 85 Mod 0, Mod 1, or Mod 2)

EQUIPMENT STATUS			
CONTRACTOR CERTIFICATION		GOVERNMENT ACCEPTANCE	
Contractor		Date of Acceptance	
Contract		Accepted by	
Equipment Ser.No.		Title	
Insp. Req. Satisfied		Agency	

FIGURE 4. SAMPLE OF APPENDIX REPORT FORM

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WS 9998

APPENDIX

REPORT FORM

QUALITY ASSURANCE PROVISION

QUALITY CONFORMANCE TEST AND EXAMINATION

INSTRUMENTATION ASSEMBLY DL 2539153 (Type I),

DL 2539071 (Type II), or DL 2539544 (Type III)

10. SCOPE

10.1 This Report Form contains data sheets to record the results of the quality conformance tests and examinations of 4.7 of this specification. When completed, with authorized certification and acceptance signatures affixed, this report form or copies thereof shall accompany the instrumentation assembly bearing the recorded equipment serial number.

20. GENERAL INSTRUCTIONS

20.1 Test sequence. The quality conformance tests and examinations of 4.7 shall be completed and the results recorded or verified in this report form prior to acceptance of the instrumentation assembly. Unless otherwise noted, the sequence of testing, recording and verification, as performed in 4.7, is mandatory.

20.2 Recording of results. Where space is provided to indicate the value of specific reading obtained, the specific reading shall be recorded if a limit or limits is given in the Limit column for the test. The initials of the individual performing the inspection shall be inserted above the value observed and recorded. If limits are not stated in the Limit column, the individual performing the test or inspection shall initial in the space provided in the Reading column, indicating that the requirement has been met satisfactorily.

30. QUALITY CONFORMANCE TEST AND EXAMINATION

	Reference	Limits	Reading
30.1 Configuration type	4.7	Type I, Type II/ Type III	_____
30.2 Test equipment	4.7.2	In calibration	_____

FIGURE 4.(CONTD) SAMPLE OF APPENDIX REPORT FORM

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WS 9998
APPENDIX

		Reference	Limits	Reading
30.3 Test procedure				
30.3.1 Continuity tests		4.7.3.1		
From	To			
J4806-A	J4806-a		0.5 ohm max	_____
J4806-K	J4805-V, T		0.5 ohm max	_____
30.4 Visual inspections				
30.4.1 Identification marking		4.7.4.1		
30.4.2 Interchangeability				
	Diameter dimension	4.7.4.2.3	11.330 in. max	_____
	Thickness dimension	4.7.4.2.4	7.250 in. max	_____
30.5 Process verification				
30.5.1 Lacing		4.7.5.2		
30.5.2 Soldering		4.7.5.3		_____
30.5.3 Cleanliness		4.7.5.4		_____
30.6 Packaging, packing, and marking inspection				_____
30.6.1 Packaging		4.7.6.1		_____
30.6.2 Packing		4.7.6.2		_____
30.6.3 Marking		4.7.6.3		_____

FIGURE 4.(CONTD) SAMPLE OF APPENDIX REPORT FORM

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Appendix

DOCUMENTS FOR USE IN DESIGN

10. SCOPE

10.1 Scope of appendix. The scope of this appendix is to identify documents known to be useful in the design of systems, equipments, and items for NAVORDSYSCOM.

20. Specifications for ordnance and ordnance equipment shall make use of by reference or incorporate the requirements of the following documents as appropriate:

20.1 Shipboard equipment.

MIL-S-901	Shock Tests, H.i. (High-impact), Shipboard Machinery, Equipment and System, Requirements for (Navy)
MIL-STD-167	Mechanical Vibrations of Shipboard Equipment
MIL-STD-1399	Interface Standard for Shipboard Systems

20.2 Electrical and electronic equipment.

a. Design practices:

MIL-STD-454	Standard General Requirements for Electronic Equipment

b. Selection of standard parts:

MIL-STD-1631	Procedure for Selection of Electronic and Electrical Parts During Equipment Design

c. Physical packaging design (for ordnance equipment rather than ordnance):

MIL-STD-1378	Requirements for Employing Standard Hardware Program Modules

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MIL-STD-1389	Design Requirements for Standard Hardware Program Electronic Modules

d. Bonding and grounding:

MIL-STD-1310	Shipboard Bonding and Grounding Methods for Electromagnetic Compatibility

20.3 Fire control equipment.

MIL-F-18870	Fire Control Equipment, Naval Ship and Shore, General Specification

20.4 Painting of equipment.

MIL-STD-1303	Painting of Naval Ordnance Equipment

20.5 Lubrication of equipment.

NAVORD OD 3000	Lubrication of Ordnance Equipment

20.6 Test equipment.

MIL-STD-1364	Preferred General Purpose Electronic Test Equipment

20.7 Selection of components/equipments.

MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-680	Contractor Standardization Plans and Management

20.8 Reliability.

MIL-STD-785	Reliability Program for Systems and Equipment Development and Production

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20.9 Maintainability.

MIL-STD-470	Maintainability Program Requirements (for Systems and Equipments)

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